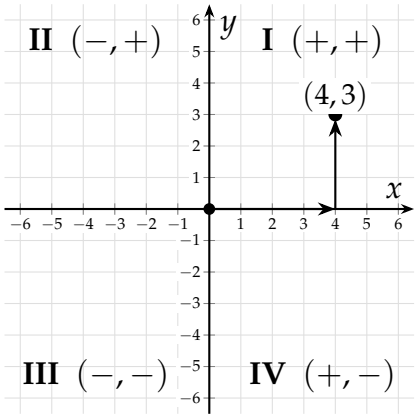
Graphing on the Coordinate Plane

Test review: points, quadrants, reflections, distance, shapes and proportional graphs

Answer Key

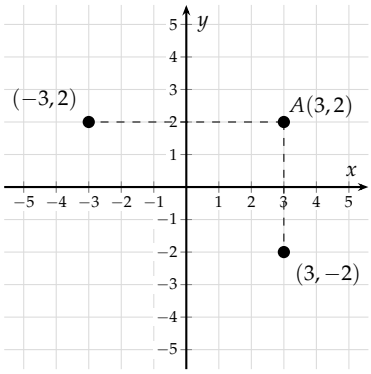
1 The coordinate plane



- The x**-axis** goes left–right; the y**-axis** goes up–down. They cross at the **origin** $(0, 0)$.
- An **ordered pair** (x, y): **first** walk left/right (x), **then** up/down (y).
  "Walk to the elevator, then ride it."
- The axes cut the plane into four **quadrants**, numbered I–IV **counterclockwise**, starting at the top right.
- A point **on an axis** (like $(0, 5)$ or $(-2, 0)$) is in **no** quadrant.

Watch out: $(4, 3)$ and $(3, 4)$ are **different** points. Order matters: x always comes first.

2 Reflections — flip over an axis



Reflect over…	Rule	Example
the x-axis	$(x, y) \to (x, -y)$	$(3, 2) \to (3, -2)$
the y-axis	$(x, y) \to (-x, y)$	$(3, 2) \to (-3, 2)$

Memory trick: over the x**-axis**, the x **stays**; only y changes sign. Over the y**-axis**, the y **stays**.

3 Distance between two points on a line

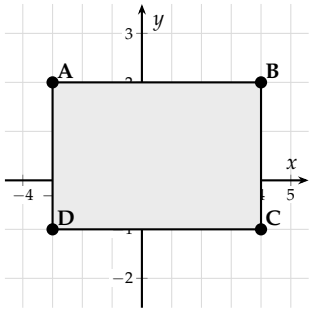
Same x or same y? Count — or use absolute value

Same quadrant / same side: subtract. $(2,5)$ to $(2,1)$: $5-1=4$ units.
Opposite sides of an axis: add the absolute values. $(-3,4)$ to $(5,4)$: $|-3|+|5| = 3+5=8$ units.
One rule for both: **distance** $=$ |difference|. $|5-(-3)| = |8| = 8$. Distance is never negative.

Simple case first: from $(-1,0)$ to $(1,0)$ — count the squares: 2. And $|-1|+|1|=2$ ✓. That's why you **add** across the axis.

4 Shapes on the coordinate plane

Example · rectangle $ABCD$ with $A(-3,2)$, $B(4,2)$, $C(4,-1)$, $D(-3,-1)$

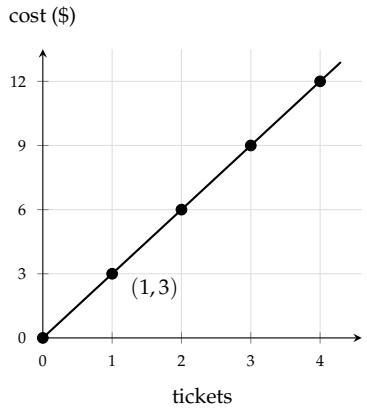


Length AB: same y, so $|4-(-3)|=7$.
Width BC: same x, so $|2-(-1)|=3$.
Perimeter $=7+3+7+3=20$ units.
Area $=7\times 3=21$ square units.

5 Graphing a proportional relationship

Example · Tickets cost \$3 each

tickets x	0	1	2	3	4
cost y (\$)	0	3	6	9	12



How to tell it's proportional from a graph:

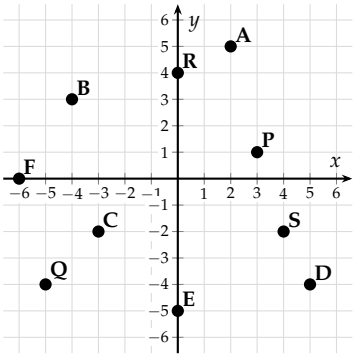
1. the points are on a **straight line**, and
2. the line goes **through the origin** $(0, 0)$.

The **constant of proportionality** (unit rate) is $k = \frac{y}{x} = \frac{3}{1} = 3$. Equation: $y = 3x$.
Look for the point $(1, k)$: here $(1, 3)$ means *1 ticket costs \$3*.
A point like $(4, 12)$ means *4 tickets cost \$12*.

Not proportional: a straight line that **doesn't** pass through $(0, 0)$ — for example a \$5 entry fee plus \$3 per ride: $y = 3x + 5$ starts at $(0, 5)$.

6 Practice

A. Read and plot points EASY



1. Write the coordinates of each point.
$A\ (2, 5)$ $B\ (-4, 3)$
$C\ (-3, -2)$ $D\ (5, -4)$
$E\ (0, -5)$ $F\ (-6, 0)$

2. Which quadrant is each of A, B, C, D in? Which points are on an axis?

> A: I, B: II, C: III, D: IV. E and F are on an axis (no quadrant).

3. Plot and label on the same grid:
$P(3, 1)$, $Q(-5, -4)$, $R(0, 4)$, $S(4, -2)$.

> Points P, Q, R, S are plotted on the grid.

B. Signs and quadrants EASY MEDIUM

4. Name the quadrant: (a) $(-7, 2)$ (b) $(6, -1)$ (c) $(-0.5, -8)$ (d) $(3\frac{1}{2}, 4)$

(a) II (b) IV (c) III (d) I

5. A point has a negative x-coordinate and a positive y-coordinate. Which quadrant is it in?

Quadrant II $(-, +)$.

6. True or false: the point $(0, -3)$ is in Quadrant IV. Explain.

False. Its x-coordinate is 0, so it lies **on the y-axis** — in no quadrant.

C. Reflections MEDIUM

7. Reflect $(5, -2)$ over the x-axis. Then reflect $(5, -2)$ over the y-axis.

Over the x-axis: $(5, 2)$. Over the y-axis: $(-5, -2)$.

8. $(-4, 6)$ is reflected over the y-axis, and then the new point is reflected over the x-axis. Where does it end up? Which quadrant?

$(-4, 6) \to (4, 6) \to (4, -6)$. Quadrant IV. (Two flips change both signs.)

9. Point M is (a, b) and point N is $(a, -b)$. Over which axis is N the reflection of M?

The x-axis (the x-coordinate stayed the same).

D. Distance MEDIUM

10. Find the distance between $(3, 7)$ and $(3, -2)$.

Same x. Opposite sides of the x-axis: $7 + 2 = 9$ units. (Or $|7 - (-2)| = 9$.)

11. Find the distance between $(-6, -1)$ and $(-2, -1)$.

Same y. Same side: $|-2 - (-6)| = |4| = 4$ units.

12. Point A is at $(-5, 4)$. Point B is 9 units to the right of A. Find B. Which quadrant is B in?

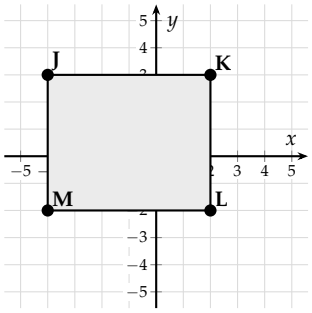
$B = (-5 + 9, 4) = (4, 4)$, Quadrant I.

13. A diver is at -12 m and a seagull is at $+15$ m, straight above the diver. How far apart are they?

$|-12| + |15| = 12 + 15 = 27$ m (opposite sides of sea level, so add).

E. Shapes MEDIUM HARD

14. Plot $J(-4,3)$, $K(2,3)$, $L(2,-2)$ and $M(-4,-2)$ and connect them in order.
(a) What shape is $JKLM$?
(b) Find its perimeter.
(c) Find its area.



(a) A rectangle. (b) $JK = |2-(-4)| = 6$, $KL = |3-(-2)| = 5$. Perimeter $= 2(6+5) = 22$ units. (c) Area $= 6 \times 5 = 30$ square units.

15. Three corners of a square are $(-1,4)$, $(3,4)$ and $(3,0)$. Find the fourth corner and the area of the square.

The fourth corner is $(-1,0)$. Side $= |3-(-1)| = 4$, so area $= 4 \times 4 = 16$ square units.

16. HARD A triangle has vertices $(-2,-3)$, $(4,-3)$ and $(1,5)$. Find its area.

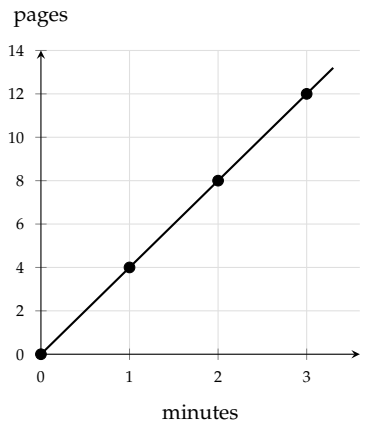
Base on $y=-3$: $|4-(-2)| = 6$. Height: from $y=-3$ up to $y=5$ is $|5-(-3)| = 8$. Area $= \frac{1}{2} \cdot 6 \cdot 8 = 24$ square units.

F. Proportional relationships MEDIUM HARD

17. A printer prints 4 pages per minute.

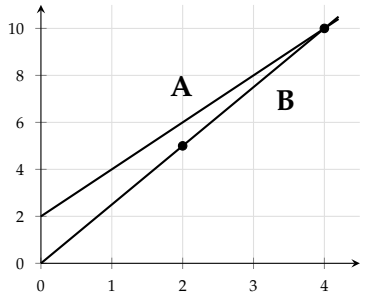
minutes x	0	1	2	3
pages y	0	4	8	12

(a) Complete the table and graph it.
(b) Is the relationship proportional? How can you tell from the graph?
(c) Write an equation. (d) What does the point $(1,4)$ mean?



(b) Yes — the points make a straight line through $(0,0)$. (c) $y = 4x$. (d) In 1 minute the printer prints 4 pages (the unit rate).

18. Which graph shows a proportional relationship? For that one, find the constant of proportionality k.



Graph **B**: a straight line through the origin, through $(2, 5)$, so $k = \frac{5}{2} = 2.5$. Graph A is a straight line but starts at $(0, 2)$, so it is **not** proportional.

19. HARD A proportional graph passes through $(6,\ 15)$. Find k, write the equation, and find y when $x = 10$.

$k = \dfrac{15}{6} = 2.5$. $y = 2.5x$. When $x = 10$: $y = 25$. (Check: $(2, 5)$ is also on the line, $5 \div 2 = 2.5$ ✓)

G. Challenge HARD

20. Point $P(-3,\ a)$ is 7 units from $Q(-3,\ 2)$. Find **both** possible values of a.

Same x, so $|a - 2| = 7$: $a = 2 + 7 = 9$ or $a = 2 - 7 = -5$. Points $(-3, 9)$ and $(-3, -5)$.

21. The rectangle $(1, 1)$, $(5, 1)$, $(5, 3)$, $(1, 3)$ is reflected over the y-axis. List the new vertices. Did the area change?

$(-1, 1)$, $(-5, 1)$, $(-5, 3)$, $(-1, 3)$. The area stays $4 \times 2 = 8$ square units — a reflection flips the shape but does not change its size.